

Your Success is our Win (58)

Put value of  $\Delta_2$  ①

$$\Delta = \Delta_1 + abc \Delta_1$$

$$\therefore \Delta = 0$$

$$0 = \Delta_1 + abc \Delta_1$$

$$\Delta_1(1+abc) = 0$$

$$1+abc = 0$$

$$\Delta_1$$

$$1+abc = 0$$

proved!

Ex#2.5

① Reduce the following matrices into echelon form using elementary row operation.

$$(i) \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$$

Sol

$$\approx \begin{bmatrix} -1 & -5 & -3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix} R_{1 \leftrightarrow 3}$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix} R_1 \times (-1)$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ -3+3(1) & 5+3(5) & 6+3(3) \\ 5-5(1) & 9-5(5) & 3-5(3) \end{bmatrix} R_2 + 3R_1, R_3 - 5R_1$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 20 & 15 \\ 0 & -16 & -12 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & (20)(\frac{1}{20}) & (15)(\frac{1}{20}) \\ 0 & -16 & -12 \end{bmatrix} (\frac{1}{20})R_2$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 3/4 \\ 0 & -16 & -12 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 3/4 \\ 0 & -16+16 & -12+16(3/4) \end{bmatrix} 16R_3 + R_2$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 3/4 \\ 0 & 0 & 0 \end{bmatrix}$$

It is an echelon form of given matrix.

$$(ii) \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{bmatrix}$$

Sol

$$\approx \begin{bmatrix} 1 & 2 & -1 \\ 2-2 & 4-2(2) & 1-2(-1) \\ 3-3 & 6-3(2) & 2-3(-1) \end{bmatrix} R_2 - 2R_1, R_3 - 3R_1$$

$$\approx \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} R_3 - R_2$$

It is an echelon form of given matrix.

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$$\text{iii)} \begin{bmatrix} 3 & -7 & 0 & 9 \\ 2 & 4 & -1 & 0 \\ 10 & 0 & -2 & -4 \end{bmatrix}$$

Sol

$$\text{ii)} \begin{bmatrix} 3-2 & -4-4 & 0+1 & 9 \\ 2 & 4 & -1 & 0 \\ 10 & 0 & -2 & -4 \end{bmatrix} R_1 - R_2$$

$$\text{ii)} \begin{bmatrix} 1 & -8 & 1 & 9 \\ 2 & 4 & -1 & 0 \\ 10 & 0 & -2 & -4 \end{bmatrix}$$

$$\text{ii)} \begin{bmatrix} 1 & -8 & 1 & 9 \\ 2-2 & 4-2(-8) & -1-2(1) & 0-2(9) \\ 10-10 & 0-10(-8) & -2-10(1) & -4-10(9) \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - 10R_1 \end{matrix}$$

$$\text{ii)} \begin{bmatrix} 1 & -8 & 1 & 9 \\ 0 & 20 & -3 & -18 \\ 0 & 80 & -12 & -94 \end{bmatrix}$$

$$\text{ii)} \begin{bmatrix} 1 & -8 & 1 & 9 \\ 0 & 20(1/20) & -3/20 & -18/20 \\ 0 & 80 & -12 & -94 \end{bmatrix} (1/20)R_2$$

$$\text{ii)} \begin{bmatrix} 1 & -8 & 1 & 9 \\ 0 & 1 & -3/20 & -9/10 \\ 0 & 80 & -12 & -94 \end{bmatrix}$$

$$\text{ii)} \begin{bmatrix} 1 & -8 & 1 & 9 \\ 0 & 1 & -3/20 & -9/10 \\ 0 & 80-80 & -12-80(-3/20) & -94-80(-9/10) \end{bmatrix} R_3 - 80R_2$$

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$$\text{ii)} \begin{bmatrix} 1 & -8 & 1 & 9 \\ 0 & 1 & -3/20 & -9/10 \\ 0 & 0 & 0 & -22 \end{bmatrix} \sim \begin{bmatrix} 1 & -8 & 1 & 9 \\ 0 & 1 & -3/20 & -9/10 \\ 0 & 0 & 0 & 1 \end{bmatrix} (-1/22)R_3$$

② Reduced the following matrices into reduced echelon form using elementary row operation.

$$\text{(i)} \begin{bmatrix} 3 & 5 & 4 \\ 4 & 1 & 5 \\ 7 & 6 & 3 \end{bmatrix}$$

Sol

$$\text{ii)} \begin{bmatrix} 3/3 & 5/3 & 4/3 \\ 4 & 1 & 5 \\ 7 & 6 & 3 \end{bmatrix} (1/3)R_1$$

$$\text{ii)} \begin{bmatrix} 1 & 5/3 & 4/3 \\ 4 & 1 & 5 \\ 7 & 6 & 3 \end{bmatrix}$$

$$\text{ii)} \begin{bmatrix} 1 & 5/3 & 4/3 \\ 4-4 & 1-4(5/3) & 5-4(4/3) \\ 7-7 & 6-7(5/3) & 3-7(4/3) \end{bmatrix} \begin{matrix} R_2 - 4R_1 \\ R_3 - 7R_1 \end{matrix}$$

$$\text{ii)} \begin{bmatrix} 1 & 5/3 & 4/3 \\ 0 & -17/3 & -1/3 \\ 0 & -17/3 & -19/3 \end{bmatrix}$$

$$\text{ii)} \begin{bmatrix} 1 & 5/3 & 4/3 \\ 0 & (-17/3)(-3/17) & (-1/3)(3/17) \\ 0 & -17/3 & -19/3 \end{bmatrix} (-3/17)R_2$$



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$$\sim \begin{bmatrix} 1 & 5/3 & 4/3 \\ 0 & 1 & 1/17 \\ 0 & -17/3 & -19/3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1+0 & 5/3-5/3 & 4/3-(5/3)(1/17) \\ 0 & 1 & 1/17 \\ 0 & -17/3 & -19/3 \end{bmatrix} R_1 - (5/3)R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 21/17 \\ 0 & 1 & 1/17 \\ 0 & -17/3 & -19/3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 21/17 \\ 0 & 1 & 1/17 \\ 0 & -17/3+17/3 & -19/3+1/17 \end{bmatrix} R_3 + (17/3)R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 21/17 \\ 0 & 1 & 1/17 \\ 0 & 0 & -6 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 21/17 \\ 0 & 1 & 1/17 \\ 0 & 0 & (-6)(-1/6) \end{bmatrix} (-1/6)R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 21/17 \\ 0 & 1 & 1/17 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 21/17-21/17 \\ 0 & 1 & 1/17 \\ 0 & 0 & 1 \end{bmatrix} R_1 - (21/17)R_3$$

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$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1/17 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1/17-1/17 \\ 0 & 0 & 1 \end{bmatrix} R_2 - 1/17 R_3$$

It is the reduced echelon form of given matrix.

$$(ii) \begin{bmatrix} 0 & 3 & -2 \\ 2 & -4 & 6 \\ 2 & 3 & -1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & -4 & 6 \\ 0 & 3 & -2 \\ 2 & 3 & -1 \end{bmatrix} R_{12}$$

$$\sim \begin{bmatrix} 2/2 & -4/2 & 6/2 \\ 0 & 3 & -2 \\ 2 & 3 & -1 \end{bmatrix} (1/2)R_1$$

$$\sim \begin{bmatrix} 1 & -2 & 3 \\ 0 & 3 & -2 \\ 2 & 3 & -1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 3 \\ 0 & 3 & -2 \\ 2-2 & 3+4 & -1-6 \end{bmatrix} R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & -2 & 3 \\ 0 & 3 & -2 \\ 0 & 7 & -7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 3 \\ 0 & 3-2 & -2+3 \\ 0 & 7 & -7 \end{bmatrix} R_2 + R_1$$

$$\sim \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 1 \\ 0 & 7 & -7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2+2 & 3+2 \\ 0 & 1 & 1 \\ 0 & 7 & -7 \end{bmatrix} R_1 + 2R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 7 & -7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 7-7 & -7-7 \end{bmatrix} R_3 - 7R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & -14 \end{bmatrix}$$

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$$\sim \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & (-14)(-\frac{1}{14}) \end{bmatrix} \xrightarrow{(-\frac{1}{14})R_3}$$

$$\sim \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 5-5 \\ 0 & 1 & 1-1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 - 5R_3 \\ R_2 - R_3 \end{matrix}}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{It is the reduced echelon form of given matrix.}$$

$$(iii) \begin{bmatrix} 1 & 1 & 1 & -1 \\ 2 & 2 & 1 & -3 \\ -1 & -1 & 1 & -3 \end{bmatrix}$$

Sol

$$\sim \begin{bmatrix} 1 & 1 & 1 & -1 \\ 2-2 & 2-2 & 1-2 & -3+2 \\ -1+1 & -1+1 & 1+1 & -3+1 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 - 2R_1 \\ R_3 + R_1 \end{matrix}}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 2 & -2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 2-2 & -4+2 \end{bmatrix} \xrightarrow{\begin{matrix} -R_2 \\ R_3 - 2R_2 \end{matrix}}$$

(65)

$$\sim \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -6 \end{bmatrix} \sim$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 - R_2 \\ \frac{1}{6}R_3 \end{matrix}}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & -2+2 \\ 0 & 0 & 1 & 1-1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 + 2R_3 \\ R_1 - R_3 \end{matrix}}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

It is the reduced row echelon form of matrix.

⑤ Find rank of the following matrices.

$$A) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 0 & 2 & 2 \end{bmatrix}$$

Sol

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 2-2 & 2-4 & 4-6 \\ 0 & 2 & 2 \end{bmatrix} \xrightarrow{R_2 - 2R_1}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 2 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 2 & 2 \end{bmatrix} \xrightarrow{-R_2}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 2-2 & 2-4 \end{bmatrix} \xrightarrow{R_3 - 2R_2}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$



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$$\cong \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & (-2)(-1/2) \end{bmatrix} R_3(-1/2)$$

$$\cong \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Rank of given matrix is 3.

$$(iii) \begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 2 \\ -1 & 2 & 2 \end{bmatrix}$$

Sol

$$\cong \begin{bmatrix} 3 & 1 & 2 \\ 2 & 3 & 4 \\ -1 & 2 & 2 \end{bmatrix} R_{12}$$

$$\cong \begin{bmatrix} 3-2 & 1-3 & 2-4 \\ 2 & 3 & 4 \\ -1 & 2 & 2 \end{bmatrix} R_1-R_2$$

$$\cong \begin{bmatrix} 1 & -2 & -2 \\ 2 & 3 & 4 \\ -1 & 2 & 2 \end{bmatrix}$$

$$\cong \begin{bmatrix} 1 & -2 & -2 \\ 2-2 & 3+4 & 4+4 \\ -1+1 & 2-2 & 2-2 \end{bmatrix} \begin{matrix} R_2-2R_1 \\ R_3+R_1 \end{matrix}$$

$$\cong \begin{bmatrix} 1 & -2 & -2 \\ 0 & 7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

Scale of given matrix is 2.

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$$(iii) \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -7 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$$

Sol

$$\cong \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -7 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix} R_{14}$$

$$\cong \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ -1+1 & -2+4 & -1+5 & 3-9 & 1-7 \\ -2+2 & -3+8 & 0+10 & 3+2(-9) & -7-14 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix} \begin{matrix} R_2+R_1 \\ R_3+2R_1 \end{matrix}$$

$$\cong \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 5 & 10 & -15 & -21 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

$$\cong \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2(1/2) & 4(1/2) & -6(1/2) & -6(1/2) \\ 0 & 5 & 10 & -15 & -21 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix} (1/2)R_2$$

$$\cong \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 5 & 10 & -15 & -21 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

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$$\begin{array}{l}
 \text{ii)} \left[ \begin{array}{ccccc} 1 & 4 & 5 & -9 & -7 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 5-5 & 10-10 & -15+15 & -21+21 \\ 0 & -3+3 & -6+6 & 9-9 & 9-9 \end{array} \right] \begin{array}{l} R_3 - 5R_2 \\ R_4 + 3R_2 \end{array}
 \end{array}$$

$$\begin{array}{l}
 \text{iii)} \left[ \begin{array}{ccccc} 1 & 4 & 5 & -9 & -7 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 0 & 0 & 0 & -6 \\ 0 & 0 & 0 & -5 & 0 \end{array} \right] \begin{array}{l} \text{Rank of given} \\ \text{matrix is 4} \end{array}
 \end{array}$$

Q Find the inverse of the following matrices using elementary row operation.

$$\text{1)} \left[ \begin{array}{ccc} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{array} \right]$$

Sol

$$\text{Let, } A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{vmatrix}$$

$$|A| = 1(4+0) - 2(0) - 3(0-4)$$

$$|A| = -4 - 0 + 12$$

$$|A| = 8$$

As  $|A| \neq 0$ . So A is non-singular matrix and its inverse exist.

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Appending Is in the right side of A.

$$= \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 & 1 & 0 \\ -2 & -2 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$\begin{array}{l}
 \text{ii)} \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 & 1 & 0 \\ -2 & -2 & 2 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \\ R_3 + 2R_1 \end{array}
 \end{array}$$

$$\begin{array}{l}
 \text{iii)} \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 & 1 & 0 \\ 0 & 2 & -4 & 2 & 0 & 1 \end{array} \right]
 \end{array}$$

$$\begin{array}{l}
 \text{iv)} \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & (-2)(-1/2) & 0 & 0 & -1/2 & 0 \\ 0 & 2 & -4 & 2 & 0 & 1 \end{array} \right] \begin{array}{l} \\ -1/2 R_2 \end{array}
 \end{array}$$

$$\begin{array}{l}
 \text{v)} \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/2 & 0 \\ 0 & 2 & -4 & 2 & 0 & 1 \end{array} \right]
 \end{array}$$

$$\begin{array}{l}
 \text{vi)} \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/2 & 0 \\ 0 & 2-2 & -4-0 & 2 & -1/2-2(-1/2) & 1 \end{array} \right] \begin{array}{l} \\ R_3 - 2R_2 \end{array}
 \end{array}$$

$$\begin{array}{l}
 \text{vii)} \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/2 & 0 \\ 0 & 0 & -4 & 2 & 1 & 1 \end{array} \right]
 \end{array}$$

$$\begin{array}{l}
 \text{viii)} \left[ \begin{array}{ccc|ccc} 1 & 2-2 & -3-0 & 1-0 & 0-2(-1/2) & 0 \\ 0 & 1 & 0 & 0 & -1/2 & 0 \\ 0 & 0 & -4 & 2 & 1 & 1 \end{array} \right] \begin{array}{l} \\ R_1 - 2R_2 \end{array}
 \end{array}$$



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$$\text{ii) } \left[ \begin{array}{cccc|c} 1 & 0 & -3 & 1 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & -4 & 2 & 1 \end{array} \right]$$

$$\text{ii) } \left[ \begin{array}{cccc|c} 1 & 0 & -3 & 1 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & -4(-\frac{1}{4}) & 2(-\frac{1}{4}) & 1(-\frac{1}{4}) \end{array} \right] (-\frac{1}{4}) R_3$$

$$\text{iii) } \left[ \begin{array}{cccc|c} 1 & 0 & -3 & 1 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & -\frac{1}{4} \end{array} \right]$$

$$\text{iii) } \left[ \begin{array}{cccc|c} 1 & 0 & -3+2 & 1+3(-\frac{1}{2}) & 0+3(-\frac{1}{4}) \\ 0 & 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & -\frac{1}{4} \end{array} \right] R_1 + R_3$$

$$\text{iii) } \left[ \begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{2} & -\frac{3}{4} \\ 0 & 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & -\frac{1}{4} \end{array} \right]$$

$$\text{iv) } \left[ \begin{array}{ccc} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 1 & 0 & 2 \end{array} \right]$$

$$\text{let, } A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 1 & 0 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 1 & 0 & 2 \end{vmatrix}$$

$$|A| = 1(-2-0) - 2(-2-0) - 1(-2-0)$$

$$|A| = 1(-2-0) - 2(-2-0) - 1(-2-0)$$

$$|A| = -2 + 4 - 1$$

$$|A| = 1$$

As  $|A| \neq 0$ ,  $\therefore$  it's inverse exist

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Appending  $I_3$  on right side of matrix A

$$= \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$\text{ii) } \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array} \right] R_3 - R_1$$

$$\text{iii) } \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 0 & -2 & 3 & -1 & 0 & 1 \end{array} \right]$$

$$\text{iii) } \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 0 & -2 & 3 & -1 & 0 & 1 \end{array} \right]$$

$$\text{iii) } \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 0 & 0 & -3 & -1 & -2 & 1 \end{array} \right] R_3 + 2R_1$$

$$\text{iii) } \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 0 & 0 & (-3)(-\frac{1}{3}) & (-1)(-\frac{1}{3}) & (-2)(-\frac{1}{3}) & (1)(-\frac{1}{3}) \end{array} \right] (-\frac{1}{3}) R_3$$

$$\text{iii) } \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{array} \right]$$

$$\text{iii) } \left[ \begin{array}{ccc|ccc} 1 & 2 & -2 & -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{array} \right] R_1 \rightarrow R_3$$

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$$\approx \begin{bmatrix} 1 & 0 & 5 & : & 1 & 2 & 0 \\ 0 & 1 & -3 & : & 0 & -1 & 0 \\ 0 & 0 & 1 & : & 1/3 & 2/3 & -1/3 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 0 & 5-5 & : & 1-5(1/3) & 2-5(2/3) & 0-5(-1/3) \\ 0 & 1 & -3+3 & : & 0+3(1/3) & -1+3(2/3) & 0+3(-1/3) \\ 0 & 0 & 1 & : & 1/3 & 2/3 & -1/3 \end{bmatrix} \begin{matrix} R_1 - 5R_3 \\ R_2 + 3R_3 \end{matrix}$$

$$\approx \begin{bmatrix} 1 & 0 & 0 & : & -2/3 & -4/3 & 5/3 \\ 0 & 1 & 0 & : & 1 & 1 & -1 \\ 0 & 0 & 1 & : & 1/3 & 2/3 & -1/3 \end{bmatrix}$$

Al

$$(iii) \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

sol

$$\text{let, } A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & -3 & 2 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$$

$$|A| = 1(1 \cdot 1 - 0 \cdot 2) + 3(2 \cdot 0 - 2 \cdot 1) + 2(2 \cdot 1 - 0 \cdot 1)$$

$$|A| = 1(1+0) + 3(2-0) + 2(-2-0)$$

$$|A| = 1(1) + 3(2) + 2(-2)$$

$$|A| = 3$$

As  $|A| \neq 0$ . So its inverse exist.

Appending  $I_3$  on right sides of matrix A.

$$= \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 2 & 1 & 0 & : & 0 & 1 & 0 \\ 0 & -1 & 1 & : & 0 & 0 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 2-2 & 1+6 & -4 & : & -2 & 1 & 0 \\ 0 & -1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \begin{matrix} \\ R_2 - 2R_1 \\ \end{matrix}$$

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$$\approx \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 0 & 1 & -4 & : & -2 & 1 & 0 \\ 0 & -1 & 1 & : & 0 & 0 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 0 & 1 & -4/7 & : & -2/7 & 1/7 & 0 \\ 0 & -1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \begin{matrix} \\ (1/7)R_2 \\ \end{matrix}$$

$$\approx \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 0 & 1 & -4/7 & : & -2/7 & 1/7 & 0 \\ 0 & -1+1 & 1+(-4/7) & : & 0 & 0 & 1 \end{bmatrix} \begin{matrix} \\ \\ R_3 + R_2 \end{matrix}$$

$$\approx \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 0 & 1 & -4/7 & : & -2/7 & 1/7 & 0 \\ 0 & 0 & 3/7 & : & -2/7 & 1/7 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 0 & 1 & -4/7 & : & -2/7 & 1/7 & 0 \\ 0 & 0 & 3/7 & : & -2/7 & 1/7 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 0 & 2/7 & : & 1/7 & 0 & 0 \\ 0 & 1 & -4/7 & : & -2/7 & 1/7 & 0 \\ 0 & 0 & 1 & : & -2/3 & 1/3 & 7/3 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 0 & 0 & : & 1/3 & 1/3 & -2/3 \\ 0 & 1 & 0 & : & -2/3 & 1/3 & 4/3 \\ 0 & 0 & 1 & : & -2/3 & 1/3 & 7/3 \end{bmatrix}$$

$$\approx A^{-1} = \begin{bmatrix} 1/3 & 1/3 & -2/3 \\ -2/3 & 1/3 & 4/3 \\ -2/3 & 1/3 & 7/3 \end{bmatrix}$$